

A standardization process for mental model analysis in socio-ecological systems

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1. Introduction

Socio-ecological system dynamics include interactions between system users and managers with biophysical elements. Often the socio-political underpinnings of a socio-ecological system are difficult to measure and incorporate into system dynamics, despite their importance in understanding risk and vulnerability in socio-ecological systems (Eakin et al., 2017). Specifically, in the case of hydrological risk, understanding the relationship between the perspectives and resultant actions of system users and managers is both important and difficult to measure. Among other important forms of participatory modeling (Aubert and Lienert, 2019; Kotir et al., 2017; Salliou et al., 2017), mental models are cognitive structures that are linked to behavior and decision-making, and therefore can be considered a “basis for people’s actions” (Jones et al., 2014). Fuzzy Cognitive Maps, a type of mental model, have been used to link how actors view the system in order to understand socio-ecological system management (Craven et al., 2017; Giabbanelli et al., 2017; Gray et al., 2014; Htun et al., 2016; Jones et al., 2014; Lynan and Brown, 2012; Özesmi and Özesmi, 2004).

Despite the value in using mental models to understand how humans visualize the dynamics of their particular socio-ecological system, there are methodological decisions that represent trade-offs in the collection of mental models. Two very clear examples of such compromises are when deciding between accuracy versus efficiency by collecting individual or group models. Another example is using models with predefined variables versus freely-defined variables or *folk variables* in which one has to decide in favor of comparability or representation (Gray et al., 2014). Each of these decisions are associated with distinct data collection and processing realities. In the case of mental models with freely-defined variables, one of the main benefits is that actors are able to freely articulate their representation of how they perceive system dynamics. However, when trying to compare mental models across actors or sectors, the diversity of variables and concepts used by actors poses a challenge for data analysis.

In this article we offer a road-map that illuminates the possibilities of standardization or streamlining mental model data with freely-defined variables for the sake of comparison. We show a step-by-step procedure -implemented in a software script- that we used in our analysis of mental models of a diversity of actors that are managers or users of the socio-ecological system of hydrological risk in Mexico City. We also offer links to code that could potentially be of use for those facing similar difficulties. Finally, we offer examples of the analysis of narratives that emerges from our standardization process.

2. Context

The analysis of mental models presented here is part of a larger project which aims to model complex socio-ecological dynamics and hydrological risk in Mexico City.¹ As part of the project, we collected 35 mental models from semi-structured interviews about the causes and consequences of water scarcity and flooding in Mexico City. The actors participating in the project included residents from three city boroughs –*Delegaciones*–, which are the main water system users, as well as local and city-level government managers. One of our main goals was to understand how narratives vary between actors and how those narratives could affect both the management of hydrological risk and the reaction to hydrological events. Therefore, each mental model contained a distinct set of variables that, in some instances, overlapped with variables used in other models, in others, were similar terms (i.e. trash and waste), and in others, were totally distinct. As a result, we developed a standardization process to be able to compare the mental models across actors. Once the mental models were standardized, we were able to identify narratives and see how they related to different actors or groups of actors, as we describe below.

3. Data collection

The interviews were based on the method proposed by Morgan et al.

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¹ For more information: <http://megadapt.weebly.com/>.

(2002) to elicit mental models. Specifically, data were collected in two different ways: 1) individual interviews and 2) workshops with groups of actors from particular neighborhoods without a list of pre-selected terms or variables. For both interviews and workshops, participants were questioned about the water situation in their neighborhoods. In the case of interviews, the conversations were recorded and used by the researchers to draw the mental models by hand on paper. The resulting mental models were validated with the participants. As for the workshop, mental models were created by the group of participants together in consensus mode. The group used as inspiration the discussions from previous group dynamics that defined different variables or concepts that were important in understanding hydrological risk. For this reason, every mental model had the variables defined by each individual or group of interviewees.

The workshops and interviews were recorded and then transcribed. These transcriptions were the raw material for the identification of stakeholders' mental models. From the different interviews, based on qualitative coding methods (Strauss and Corbin, 1998; Russell, 2017), ideas were identified and then each of these were condensed in a single variable. For example, some would talk about "water supply", "water supply using the hydric network", "water supply from a water well", or "water supply using pipelines". For all these cases, a standard variable could be "water supply". The mental model was formed by connecting a cause and effect relationship between the different variables. Pairs of variables and their concepts conformed the components of the mental models.

In order to graphically represent actors' mental models, each set of elicited pairs of variables was structured as a connectivity map or edgelist (i.e., a list of paired variables separated by commas as in: "water quality, tourism"). Although there are different graph representation data formats such as adjacency matrices and adjacency lists, we used edgelists because they are very intuitive –it is just a list of pairs that reflect the cause and effect relationships that we recorded. We also used edgelists for its simplicity in analyzing its content using either Cytoscape (Shannon et al., 2003), a complex network analysis software, or Networkx (Hagberg et al., 2008), a Python language library for the creation, manipulation, and study of complex networks. Although every mental model was visualized in a graph and we could perform different analyses to each of them, no comparison could be made if every mental model was different in terms of their variables, hence we had to standardize them.

4. Variables standardization

4.1. Variable equivalences and dictionary

It is not uncommon to find in the literature that researchers collect mental models without a set of predefined variables –sometimes a couple of variables are offered as seeds to initiate the process (Özesmi, 2006; Jones et al., 2014; Gray et al., 2015). Those who later in their project become interested in comparing different mental models are faced with a similar situation as the one we are describing here. The process of standardizing variables can be a time-consuming and exhausting task if performed by hand and even more importantly, prone to many mistakes, especially when working with many mental models and a large number of variables. In order to avoid these types of challenges we designed a human-computer strategy to reduce the minimum human errors and accelerate the process of creating a new set of standardized mental models (Fig. 1).

In our particular case, we collected 35 mental models which in total added up to 150 variables. The first task towards standardization was to create a table of equivalences among variables. This is a job that has to be performed by a human, preferably someone who was involved in the workshops, interviews, transcription or in the process of abstracting ideas into variables by a process of qualitative codification independently performed by several members of the team. The point is to

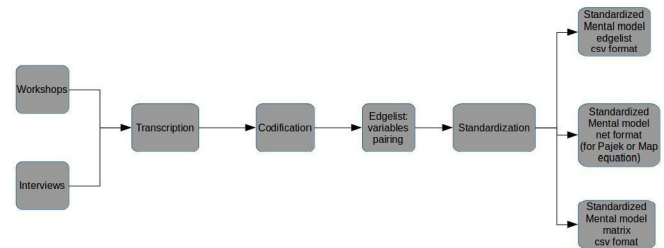


Fig. 1. General mental model data-processing pipeline.

have someone who knows the context and content for creating equivalences, because it is not merely a process of identifying synonyms, but rather a process which creates semantic equivalences.

The table of equivalences may be the most relevant step in performing comparisons that are meaningful. This table is used to create what we call a "dictionary of terms". As in any other dictionary we defined *key* terms and *value* terms, which in our case had a structure of a *folk variable* for a *standardized variable*. This is the structure we followed to create a file that included every equivalence relationship. In technical terms, such a file had a *json*² format that could be given as an input to a script that would use it to create every equivalence for every mental model.

4.2. The script

The script or program is a fundamental component that performs the work that may be time-consuming and prone to errors when conducted by humans. The script we developed receives two different inputs: a) a dictionary of variables equivalences and b) the original mental models with folk variables. The mental models are given one by one to the script but this can also be automated. The dictionary, as already mentioned, is in a *json* format. To feed the mental models to the script, these must be structured as edgelists (pairs of variables) and in a *txt* or *plain text* format or, preferably in a *comma separated values* format.

Once both the files are given to the script, it performs a task in which it looks at each folk variable in a mental model, it searches for it in the dictionary and when it finds it, in a new file it substitutes the folk variable for its standardized version, maintaining the structure of the edgelist. We are aware that the edgelist is not a universal structure for networks and mental model analysis. For this reason we decided that the output of the script would include a *csv* edgelist that can be used in different network analysis software, a *matrix*, and a *net* format that is popular for Pajek (Batagelj et al., 1998) users as well as for those using *Map Equation* (Rosvall et al., 2009) for community analysis (Fig. 2).

4.3. Analyses with standardized mental models: the case-study of hydrological risk in Mexico city

Understanding local narratives is important because they contain individual and group discourses that justify and explain their behavior (Alexandridis et al., 2018; Hämäläinen, 2015; Jones et al., 2014; Singer et al., 2017). They may also be informative about the actors' perspectives about certain issues, including what is important and what is valuable to them. Actors may have a repertoire of discourses that they selectively elicit depending with whom they want to present themselves, that is, discourse can be tool available to the actor to intentionally convey a specific message and an impression about themselves to another actor. In this sense, narratives are context-dependent and are elicited under certain conditions where they are intended for a

² JSON is an open data format that is language independent and a data interchange between languages. Its structure is based on name-value, key-value pairs such as in a dictionary, i.e., concept-meaning description.

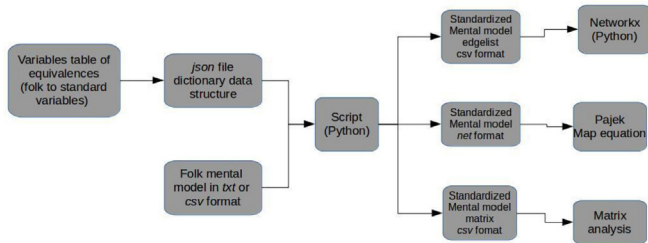


Fig. 2. Standardization pipeline and outputs.

specific audience with a specific purpose. For example, a group of residents in our project shared an ecological narrative, and they also seek government support in claiming their land titles. This dynamic shows that narratives can develop for many purposes, including to support certain needs of actors. Therefore, identifying different narratives and their components regarding a certain topic is paramount when studying socio-ecological problems because it brings to light the social and argumentative positions of each actor or group of actors involved. Moreover, it may also help to identify alliances, historical or potential, as the one between residents and government in the example above.

For our project we wanted to identify such narratives using mental models and for which it was necessary to standardize each individual model. More precisely, we sought to use the conceptual universe contained in the mental models to visualize the different narratives around water issues in an area of Mexico City and how different actors take conceptual elements from different narratives to create their own. We had three case-studies: Magdalena Contreras, Xochimilco and Iztapalapa boroughs (*delegaciones*). The mental models were standardized as has already been described. With every standardized mental model, we added them up into a single “meta-mental model”. When adding or superimposing every mental model into a one single mental model it is possible to see in a single image where there are coincidences between shared variables and patterns of connectivity among variables. This by itself is something useful to detect shared ideas and which are those that are more prevalent among actors. For example, there could be a high degree of agreement among actors on the idea that “ecological degradation” is linked to a “low community commitment to manage natural resources.”

Because our aim was to detect different narratives, we applied a community analysis to the meta-mental model. Community analysis is a way to identify groups of nodes in a network. In our case, we carried out the analysis using *Map Equation* (Rosvall et al., 2009), which informed us about which variables group together. Such groups or communities were considered the core of possible narratives emerging from the mental models seen together. The final step of this strategy was to see how many of these communities are contained in the individual mental models. This was a process that involved extracting the intersection between the set of variables of an actor and the set of variables of a particular narrative community. The similarity between any two sets of variables can be calculated by obtaining the *Jaccard Index*. As shown in Fig. 4, all that is required is to divide the intersection of the two sets over the union of them. The resulting number is between 0 and 1. Furthermore, since two actors may share the same narrative but for different reasons, for example the variables in the mental model of two actors can be part of one narrative but the variables of one actor are different from the other, we could apply the same strategy to see what are the variables and to what degree two actors sharing the same narrative converge. By doing this, we determined which actors share certain narratives and we also observed which actors have richer and more diverse perspectives, since their mental model contained a wider spectrum of narratives, and whose mental model contained only one narrative.

We show an example of a network of seven interrelated narratives in Fig. 3. Each narrative is made up of a set of variables that are strongly

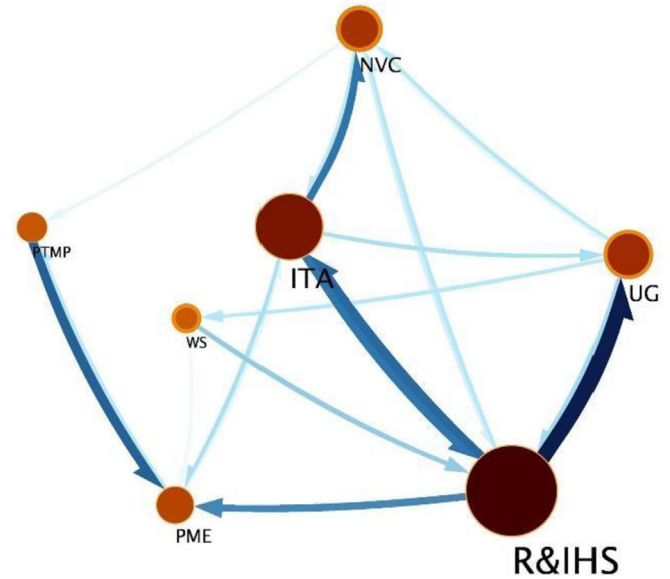


Fig. 3. Network of narratives (communities of variables). R&IHS stands for Regular and Irregular Human Settlements; ITA for Intensity of Touristic Activities; UG for urban Growth; PME for (Government) Projects Management Efficiency; NVC for Natural Vegetation Coverage; WS for Water Supply; and PTMP for Public Transportation Mobility Problems. Each name describes a set of variables that together represent elements of a narrative. The name of the community is the most central variable in the community.

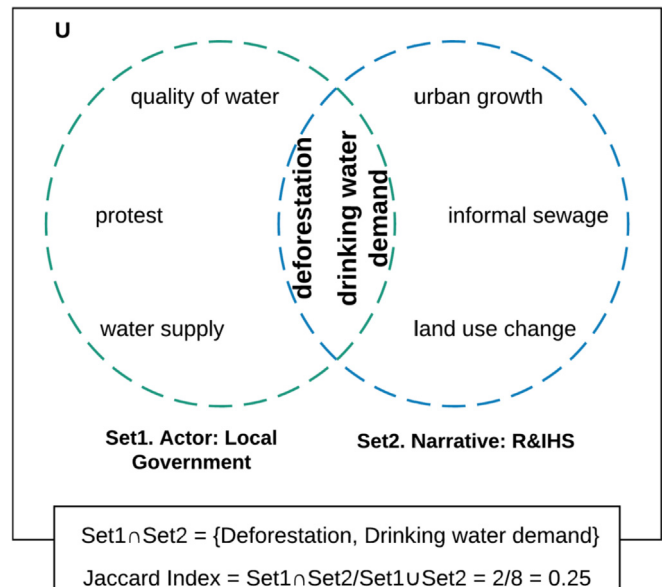


Fig. 4. An example of how to determine which variables were shared by an actor and a narrative. If the variables of a mental model and the variables of the narrative are taken as two different sets, then the intersection between them is what constitutes the shared variables. The same strategy can be applied to compare the common variables between two actors, or to compare two actors who may identify with the same narrative, although through different variables.

intertwined in a community -i.e., it conforms a network by itself. This is interpreted as each community representing an elaborated and complex idea regarding a particular topic. Such a complex idea is what we recognize as the foundation of a narrative which can be elicited in different ways in the discourse. For example, the “Regular and Irregular Human Settlements” (R&IHS) narrative is by far the largest narrative and is comprised of 20 variables. By examining the network of variables

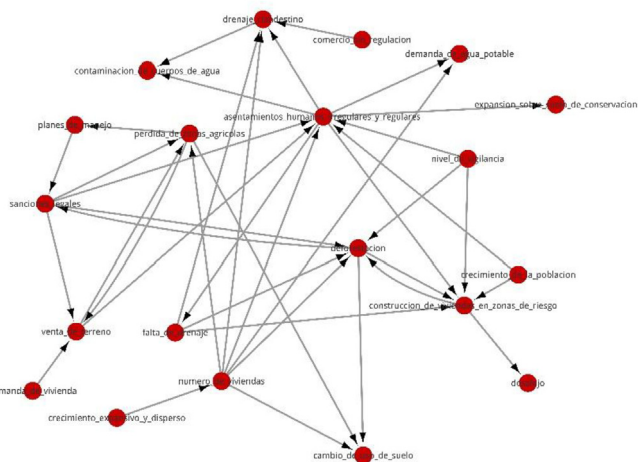


Fig. 5. The network of variables conforming the R&IHS (Regular and Irregular Human Settlements) narrative.

of the R&IHS narrative -see Fig. 5-, one can recognize that it is based on the idea that human settlements are deeply connected to ecosystem degradation. Specifically, it reflects the interviewees' understanding that urban growth is having a severe impact on deforestation, land-use change (from forest or agricultural to urban), the pollution of water bodies, and water system needs. Moreover, the problems extend to housing demand, eviction, and the lack of an efficient surveillance system to protect the area. Once we created this narrative as an output, we could analyze which actors shared part or all of this narrative in their mental models.

Identifying the presence of a narrative and to what degree in each mental model is revealing as well. As mentioned above, actors may relate to different narratives, and being able to use a narrative in the right context may help them achieve their goals. For example, for the example narrative above, we found that five of twelve interviewed actors include "Regular and Irregular Human Settlements" as one important narrative in their vision. Three of these actors represent local government offices. It becomes more revealing when the other two actors are considered. These two actors are the settlers of the two areas related to agriculture and forested land, that is, the natural areas that have been under strong pressure due to the presence of new human settlements. In the end, what we can do with this analysis is see how different actors or groups of actors overlap in terms of the narratives that they employ to discuss the issue of hydrological risk.

5. Conclusions

Understanding the underlying socio-political fabric of socio-ecological systems is challenging and exciting. Specifically, the use of mental models can help orient conflicts in resource management or ways in which certain actors overlap (or not) in their perception of system dynamics. Here we present a guide to processing mental model data when mental models have freely-defined or folk variables that make it difficult to compare across actors. Although the process still implies human work, we automated several steps in order to reduce the workload and reduce human errors. In the end, our standardization process of mental models allowed us to develop groups of narratives and analyze which actors incorporate different narratives into their mental models, which we illustrate using the example of hydrological risk in Mexico City.

Acknowledgements

This work was supported by the National Science Foundation under

Grant No. 1414052, CNH: The Dynamics of Multi-Scalar Adaptation in Megacities, with additional support from the Inter-American Institute for Global Change Research under Collaborative Research Network-CRN3: "Coping with hydrological risk in megacities: Collaborative planning framework for the Mexico City Metropolitan Area" (Project number: CRN3108). We also acknowledge the contributions of Lakshmi Charli-Joseph and Alejandra Martínez-Canedo with regards to the data collection and analysis. We are also grateful to the interviewees for their participation. Finally, we appreciate the comments of the anonymous reviewers.

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